

Causal Behavioral State Reasoning for Cyber Threat Inference in Provenance Graphs

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Problem

Advanced Persistent Threats (APTs) silently infiltrate enterprise systems for months before detection.

Real-world examples:

- **SolarWinds (2020):** Attackers lurked inside US government networks for **9 months** undetected, using legitimate software update channels
- **Target Breach (2013):** Credentials stolen via a third-party HVAC vendor; **40 million** payment cards compromised before any alert fired
- **Operation Aurora (2009):** Nation-state actors maintained persistent access to Google, Adobe, and 30+ companies using zero-day exploits disguised as normal traffic
- **Equifax (2017):** Attackers moved laterally across **48 databases** over 76 days — all activity looked like routine system behavior

Core challenge: APT actions individually look benign — the threat only emerges from the *pattern* across time and systems.

Problem Statement

How do we detect APTs on real enterprise systems without labeled data or a clean baseline?

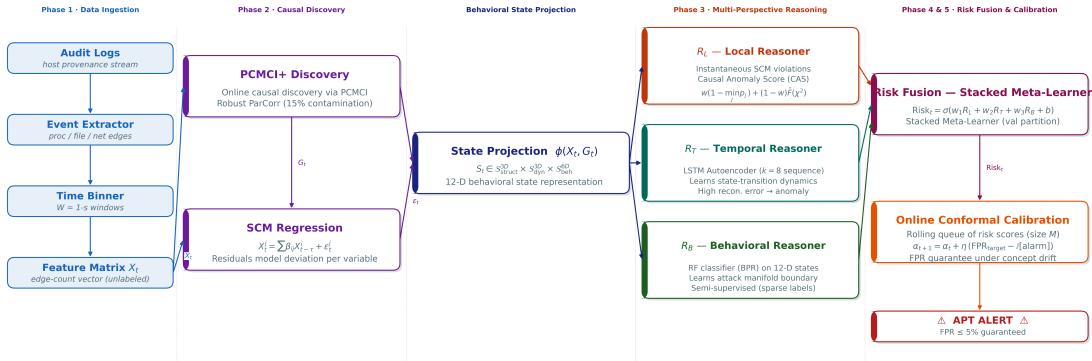
- Enterprise audit logs are **unlabeled** — security teams cannot manually inspect millions of events per day
- Attackers blend into normal operations; **individual actions are indistinguishable** from legitimate system behavior
- Systems change over time (software updates, new services) — a model trained today **drifts out of sync** tomorrow
- Waiting to collect a “clean” baseline period is unrealistic — **attackers may already be present**

Problem

Detecting stealthy, multi-stage cyber threats from **raw, uncurated, unlabeled** system logs — in real time — without assuming any clean baseline.

System Architecture

Causal Behavioral State Reasoning — End-to-End Architecture



Related Work

System	Year	Type	Key Limitation
UNICORN	2020	Unsupervised	Requires clean, attack-free training period
WATSON	2020	Unsupervised	Static profiles, no concept drift adaptation
Kilo	2022	Unsupervised	Needs uncontaminated pre-collection phase
ShadeWatcher	2022	Semi-sup.	Clean baseline + user-provided labels
Causal-IDS	2026	Semi-sup.	Static SCM fitted on benign-only logs
CausalGraph	2026	Semi-sup.	LLM inference too slow for real-time streams
OCR-APT	2025	Supervised	24–48 hrs clean pre-collection required
TraceCluster	2026	Supervised	Fully labeled subgraph clusters needed
StageFinder	2026	Supervised	12% FPR under adversarial noise

Common Gap

Every prior system assumes either a **clean baseline** or **labeled data** — neither holds in real enterprise deployments.

Proposed Solution

Core Idea

Project raw audit logs into a **Behavioral State Space** via online causal structure learning — reason across three perspectives simultaneously.

Key Insight

Causal relationships are more stable than statistical correlations — robust to noise, drift, and contamination.

Four-Layer Pipeline

- 1 **Causal Discovery** — online SCM from raw unlabeled streams
- 2 **State Reasoning** — three parallel operators:
 - R_L : instantaneous mechanism violations
 - R_T : LSTM over state trajectories
 - R_B : Random Forest on behavioral manifold
- 3 **Risk Fusion** — meta-learner combines R_L, R_T, R_B
- 4 **Conformal Calibration** — bounded FPR guarantee

Detects APTs from **uncurated, unlabeled streams** — no clean baseline required.

What is Behavioral State Space?

Step 1: Online Causal Structure Learning

Raw audit logs are binned into time windows. The **PCMCI algorithm** discovers causal links between system variables on-the-fly — no labeled data needed.

Each variable is modeled as:

$$X_t^j = \sum_{\text{parents}} \beta_{i,j} X_{t-\tau}^i + \epsilon_t^j$$

The residual ϵ_t^j measures *how much the system deviates from its causal norm*.

Step 2: State Projection

SCM residuals are compressed into a **12-dimensional state vector** S_t :

Dimension	What it captures
Structural	New / unusual graph edges
Dynamic	Activity bursts, z-scores
Behavioral	Entropy, spike ratio
Mechanism	Causal violations, χ^2
Intervention	Propagation locality

Each log window becomes a **point in state space** — attacks push the point far from the normal region.

Step 3: Multi-Perspective State Reasoning

The 12-dimensional state vector S_t is evaluated by **three parallel reasoning operators**:

R_L — Local Reasoning

Measures **instantaneous** causal mechanism violations.

Combines residual z-scores and empirical CDF of squared errors.

Catches sudden, sharp anomalies.

R_T — Temporal Reasoning

Feeds a rolling window of $k=8$ consecutive states into an **LSTM Autoencoder**.

High reconstruction error = abnormal state trajectory.

Catches slow, multi-stage APT progression.

R_B — Behavioral Reasoning

A **Random Forest** trained on a small labeled validation slice learns the boundary between normal and attack states.

Catches complex, non-linear attack patterns.

Together the three operators cover **instant shocks**, **slow drift**, and **behavioral outliers** — no single perspective can catch all APT types alone.

Step 4 & 5: Risk Fusion and Conformal Calibration

Risk Fusion

Outputs of R_L , R_T , R_B are combined by a **stacked meta-learner**:

$$\text{Risk}_t = \sigma(w_1 R_L + w_2 R_T + w_3 R_B + b)$$

Weights w_1, w_2, w_3 are trained on the validation partition to maximize AUC — the model learns *which perspective matters most* for each environment.

Conformal Calibration

Maps Risk_t to a **statistically bounded alert decision**:

- Maintains a rolling queue of past risk scores
- Computes a conformal p-value via binary search
- Raises alarm if p-value $< \alpha_t$
- α_t adapts online:

$$\alpha_{t+1} = \alpha_t + \eta (\text{target FPR} - \mathbb{I}[\text{alarm}])$$

Guarantees the false positive rate stays within the user-defined budget even under concept drift.

Results & Comparison

Head-to-head (AUC-ROC, same datasets):

Model	TC3	NODLINK	BETH	OpTC	StreamSpot
Ours (CBSR)	1.000	1.000	0.999	0.971	0.730
Pure Causal (SCM)	0.835	0.826	0.966	0.836	0.499
PyTorch Autoencoder	1.000	1.000	0.995	0.936	0.777
Isolation Forest	0.874	0.890	0.998	0.597	0.643
LOF	0.984	0.991	0.901	0.771	0.276
One-Class SVM	0.959	0.952	0.988	0.732	0.531

vs. Related Work (self-reported on own datasets[†]):

System	Metric	Score	Limitation
OCR-APT (2025)	AUC (DARPA Trace)	0.910	24–48 hrs clean pre-collection
TraceCluster (2026)	Avg. Precision	0.880	Fully labeled GNN training
StageFinder (2026)	Recall / FPR	94% / 12%	High FPR under adversarial noise
Causal-IDS (2026)	AUC (network flows)	~0.84	Static SCM, benign-only logs

[†]Direct AUC comparison not possible — different datasets and evaluation protocols.

Conclusion

- Proposed **Causal Behavioral State Reasoning (CBSR)** — a novel framework that detects APTs from raw, unlabeled, uncured enterprise logs with no clean baseline required
- Maps telemetry into a **12-dimensional behavioral state space** via online causal structure learning, evaluated in parallel by three reasoning operators (R_L , R_T , R_B)
- Achieves **AUC 0.971** on real Windows enterprise logs (OpTC), **0.999** on Linux telemetry (BETH), and **perfect 1.000** on simulated APT datasets (TC3, NODLINK)
- Resolves the pure causal model failure on heterogeneous graphs — StreamSpot improves from **0.499** $\rightarrow$ **0.730**
- Online conformal calibration controls empirical FPR within **1.78%–6.13%** under a 5% target budget
- Demonstrates a drift-adaptive, contamination-resilient, and operationally feasible pipeline for streaming enterprise security

① Robust Conformal Calibration under Contamination

Develop trimmed eCDFs and self-supervised anomaly filtering to protect the conformal calibration queue when training streams are actively poisoned by attackers

② Graph-Type Conditioning & Multi-SCM Ensembles

Explore graph-type conditioning pre-passes or multi-SCM mixture models to further handle highly heterogeneous provenance graph databases (e.g., StreamSpot-class datasets)

③ Edge Device Deployment

Optimize the SCM state projection and lightweight LSTM-AE inference to run as a low-footprint daemon on resource-constrained IoT and edge devices, pushing real-time anomaly scoring to the endpoint

Thank You

Questions & Discussion

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